



HTML Structure and Core Tags

Learn to structure the web with HTML.

ABSTRACT

HTML is the foundation of web development, used to structure and present content on the web. This module introduces the basics of HTML page structure, core tags, and editors, followed by semantic and non-semantic elements, document structure, layout, text, and formatting tags. Learners will gain a strong foundation for creating well-structured and meaningful web pages.

❖ Basics of HTML

- ✓ HTML stands for **Hyper Text Markup Language**
- ✓ It is used to design the web pages.
- ✓ With the help of HTML, you can create a complete website structure.
- ✓ HTML is the combination of Hypertext and Markup language. Hypertext defines the link between the web pages and markup language defines the text document within the tag that define the structure of web pages.
- ✓ HTML consists of a series of elements
- ✓ HTML elements tell the browser how to display the content

HTML Tags: Tags are the starting and ending parts of an HTML element. They begin with < symbol and end with > symbol. Whatever written inside < and > are called tags.

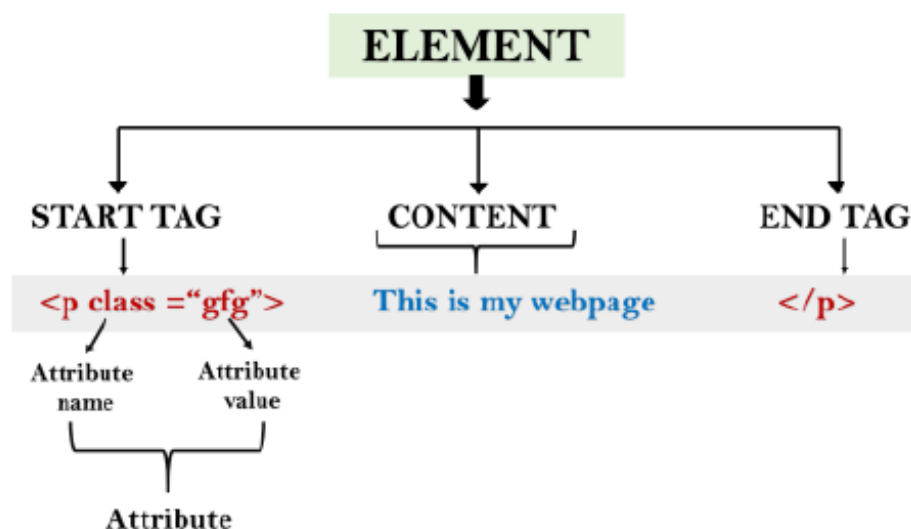
HTML element content: Elements enclose the contents in between the tags. They consist of some kind of structure or expression. It generally consists of a start tag, content and an end tag.

Example: This is the content.

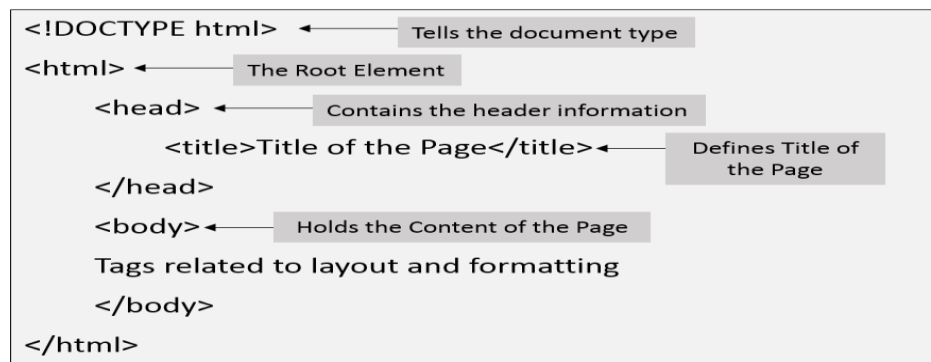
Where, is the starting tag and is the ending tag.

HTML Attributes: It is used to define the character of an HTML element. It always placed in the opening tag of an element. It generally provides additional styling (attribute) to the element.

Syntax: <tag name attribute_name= " attr_value"> content </ tag name>

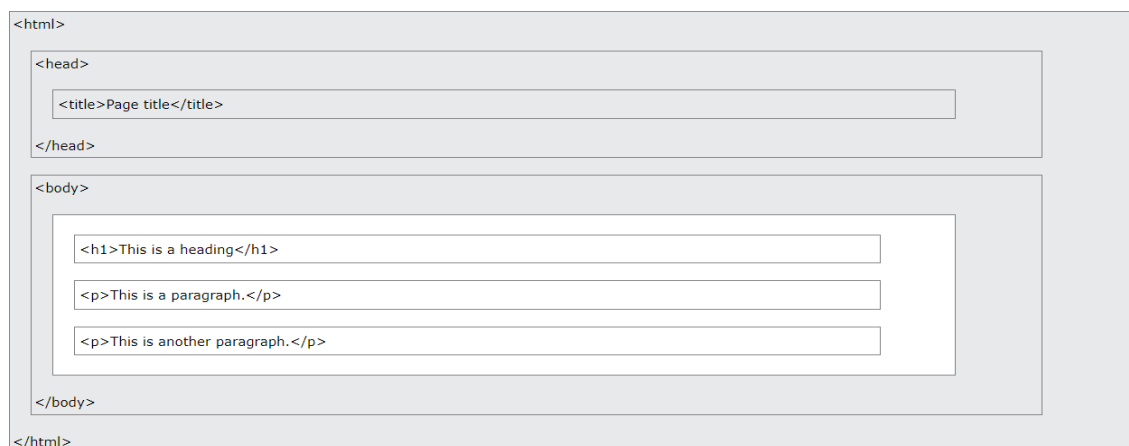


❖ Page Structure



❖ Building Blocks of HTML

1. **<!DOCTYPE html>** – A doctype or document type declaration is an instruction that tells the web browser about the markup language in which the current page is written. The doctype declaration is not case-sensitive. The declaration is not an HTML tag. It is an "information" to the browser about what document type to expect.
2. **<html>** – This tag is used to define the root element of HTML document. This tag tells the browser that it is an HTML document.
3. **<head>** – This tag is used to define the head portion of the HTML document that contains information related to the document. Elements within the head tag are not visible on the front-end of a webpage.
4. **<title>** -- This element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
5. **<body>** – The body tag is used to enclose all the visible content of a webpage. In other words, the body content is what the browser will show on the front end. For example, headings, paragraphs, images, hyperlinks, tables, lists, etc.



❖ Editors

Web pages can be created and modified by using editors (sublime, notepad++, vs code etc). However, for learning HTML it is recommended to use simple text editor like Notepad/TextEdit (Mac). It is good way to learn HTML.

First Example:

❖ Write HTML

```
<!DOCTYPE html>
<head><title>Test</title></head>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

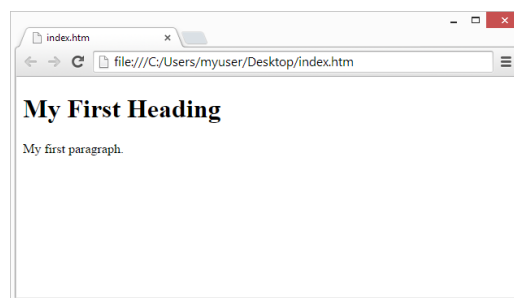
❖ Save the HTML Page

Save the file on your computer. Select **File > Save as** in the Notepad menu. Name the file **"index.html"** or **"index.htm"**

❖ View the HTML Page in Your Browser

Open the saved HTML file in your favorite browser (double click on the file, or right-click - and choose "Open with").

The result will look much like this:



❖ HTML Comments

- In HTML, comments are written using `<!--` and `-->`.
- The browser ignores everything inside the comment.
- Comments are useful to explain code, organize sections, or hide code for testing

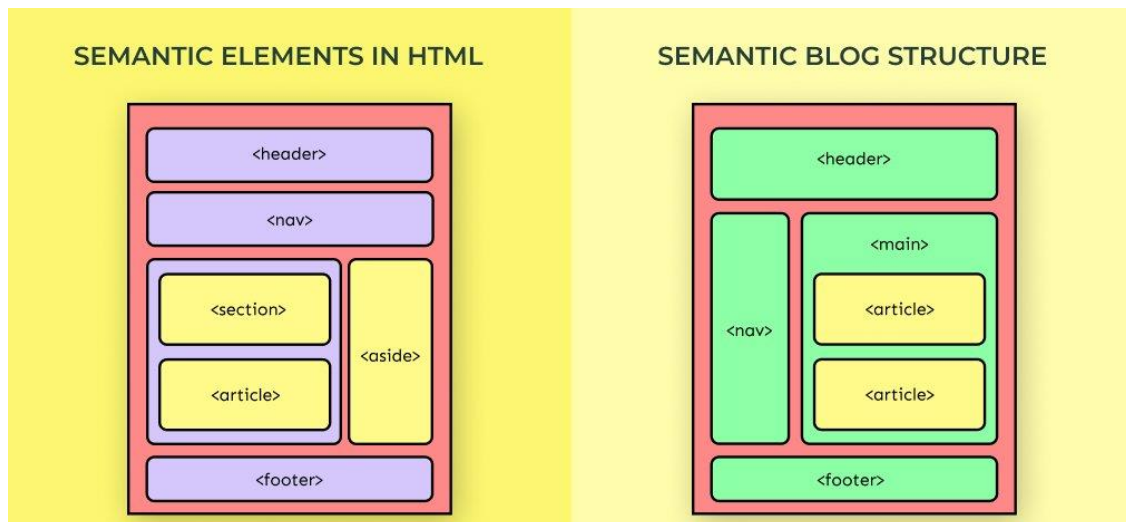
Syntax:

```
<!-- This is a comment -->
```

Example:

```
<html>
<head>
  <!-- This is the head section -->
  <title>My First Webpage</title>
</head>
<body>
  <!-- Main heading -->
  <h1>Welcome</h1>
</body>
</html>
```

❖ Semantic and Non-Semantic HTML Tags



- **Semantic Tags** → Clearly describe their meaning and purpose both to the browser and developer. They make code **more readable, SEO-friendly, and accessible**.
 - Examples:
 - `<header>` → defines header section
 - `<footer>` → defines footer section
 - `<article>` → represents an independent article
 - `<section>` → defines a thematic section
 - `<aside>` → tangential/side content (ads, related links, tips)
 - `<nav>` → navigation links
 - `<main>` → main content area
- **Non-Semantic Tags** → do not describe their meaning. They are used mainly for styling or grouping. They don't tell what the content is, only **how it should look**.
 - Examples:
 - `<div>` → division/container
 - `<span>` → inline container for text

❖ Block and Inline elements

Block Elements

Block elements always start on a new line and stretch to the full width of their parent container. They are used to structure the main layout of a page.

`<div>`, `<p>`, `<h1>` to `<h6>`, `<section>`, `<article>`, `<aside>`, `<header>`, `<footer>`, `<table>`, `<ul>`, `<ol>`, `<li>`, `<form>`, `<main>`, `<nav>`

Inline Elements

Inline elements do not start on a new line. They only take up as much width as their content requires. They are often used for formatting, styling, or embedding small chunks of content within block elements.

`<a>`, `<span>`, `<b>`, `<strong>`, `<i>`, `<em>`, `<img>`, `<br>`, `<label>`, `<input>`, `<textarea>`, `<select>`, `<button>`, `<sub>`, `<sup>`, `<abbr>`

Closing tags vs. self-closing tags

Closing tags	Self-closing tags
Closing tags necessitate the use of a closing tag. For example, tag always need	Self-closing tags do not necessitate a closing tag. For example, <input>, ,<hr>,,<meta>, < link> etc are a self-closing tag.
Closing tags include content. For example, This is a heading	Self-closing tags do not contain any content. They are either empty or receive data from attributes to render.
Child elements exist in closing tags.	Child elements do not exist in self-closing tags.

Child tags of <head> Element

- <title> → Defines the title of the web page, which appears in the browser's title bar or tab.
- <link> → Connects external resources (CSS, favicon).
- <script> → Adds JavaScript for interactivity.
- <style> → Adds CSS for styling.
- <meta> → It provides metadata (information about the webpage) for browsers and search engines.

```
<!DOCTYPE html>
<html>
<head>
  <title>Head Tags Example</title>
  <!-- Link tag: external CSS file and favicon -->

  <link rel="stylesheet" href="styles.css">

  <link rel="icon" type="image/png" href="favicon.png">

  <!-- Style tag: internal CSS -->
  <style>h1 { color: blue; }</style>

  <!-- Script tag: external JavaScript -->
  <script src="s1.js"></script>
</head>

<body><h1>Head Tags Demonstration</h1></body>
</html>
```

❖ favicon

A **favicon** (short for “**favorite icon**”) is the **small icon that represents a website**.

You’ve definitely seen them — they appear in places like:

- The **tab bar** of your browser (next to the page title).
- On **Gmail**, you see the little **red-yellow-white-green-blue envelope “M”** in the browser tab.
- On **YouTube**, you see the **red play button triangle**.
- On WhatsApp, Green speech bubble with a white telephone icon inside.

```
<link rel="icon" type="image/png" href="favicon.png">
```

- **16×16 px** → classic browser tab icon.
- **32×32, 48×48 px** → for high-res screens.
- **180×180 px** → for iOS “Add to Home Screen”.

Meta Tags in HTML

The <meta> tag is placed inside the <head> section of an HTML page.

1. Content

Content is text-based value which specifies the value associated with the **http-equiv** or **name** attribute & charset Specifies the character encoding for the HTML document.

```
<meta content="text/html">
```

2. charset

Purpose: Defines character encoding (commonly **UTF-8** – Unicode Transformation Format-8).

Real-world scenario: Ensures that special characters, accented letters, emojis, or non-English text display correctly.

```
<meta content="text/html" charset="UTF-8">
```

Examples:

- **Hindi news website:** The Times of India or BBC Hindi
 - Headline: भारत में नई शिक्षा नीति लागू
- **Gujarati news website:** Divya Bhaskar or Sandesh
 - Headline: અમદાવાદમાં નવાં વર્ષ માટે તહેવારની તૈયારી

Without UTF-8 (If browser can't predict):

Text may appear as **random symbols or question marks**.

Example:

- Wrong: ??????????????????
- Correct with UTF-8: અમદાવાદમાં નવાં વર્ષ માટે તહેવારની તૈયારી

3. name

Specifies a name for the metadata

```
<meta name="value">
```

Values for name attributes

a. viewport

Purpose: Makes a website responsive on mobile devices.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Example: Flipkart or Myntra: On a mobile phone, the layout adjusts so product images, buttons, and menus fit perfectly, making online shopping easy on smartphones.

b. description

Purpose: Short summary of the page, used by search engines.

```
<meta name="description" content="Learn the best street food recipes from India including chaat, vada pav, and golgappa.">
```

Example: When someone searches “Indian street food recipes,” Google may show this description under the page title on **Sanjeev Kapoor’s recipe website** or **Tarla Dalal recipes**.

c. keywords

Purpose: List of keywords relevant to the page content.

```
<meta name="keywords" content="Indian recipes, street food, vada pav, chaat, golgappa">
```

Example: Helps search engines understand the page topic. For instance, a blog about **Indian street food** will rank better when users search for “vada pav recipe” or “chaat recipes.”

d. author

Purpose: Indicates the creator of the webpage.

```
<meta name="author" content="Arjun Mehta">
```

Example: A food blog post written by **Arjun Mehta**, an Indian food blogger. Some browsers or CMS platforms may display the author’s name, giving proper credit.

3. http-equiv

Provides an HTTP header for the information/value of the content attribute.

```
<meta http-equiv="content-security-policy|content-type|default-style|refresh">
```

❖ **content-security-policy**

Specifies a content policy for the document.

Example:

```
<meta http-equiv="content-security-policy" content="default-src 'self'">
```

❖ **content-type**

Specifies the character encoding for the document.

Example:

```
<meta http-equiv="content-type" content="text/html; charset=UTF-8">
```

❖ **default-style**

Specified the preferred style sheet to use.

Example:

```
<meta http-equiv="default-style" content="the document's preferred stylesheet">
```

Note: The value of the content attribute above must match the value of the title attribute on a link element in the same document, or it must match the value of the title attribute on a style element in the same document.

❖ **refresh**

Provides HTTP header-like info. Can be used for refresh, content type, or caching.

```
<meta http-equiv="refresh" content="10;url=https://indiatimes.com">
```

Example: A page on **IndiaTimes** that says “Thank you for subscribing to our newsletter” and then redirects to the homepage after 10 seconds.

```
<head>
  <!-- Character encoding for Indian languages -->
  <meta charset="UTF-8">
  <!-- Responsive design for mobile devices -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <!-- Short description for search engines -->
  <meta name="description" content="Explore the best Indian street food recipes including chaat, vada pav,
pani puri, and more.">
  <!-- Relevant keywords -->
  <meta name="keywords" content="Indian recipes, street food, vada pav, chaat, pani puri, Mumbai food">
  <!-- Author of the webpage -->
  <meta name="author" content="Arjun Mehta">
  <!-- Page refresh (redirects after 30 seconds) -->
  <meta http-equiv="refresh" content="10;url=https://en.wikipedia.org/wiki/Meta_element">
</head>
```

Child tags of <body> Element

❖ Paragraph: <p>, <pre>

- **<p> Tag:** Defines a **paragraph**. Browsers automatically add a single blank line before and after each <p> element. (Lines and Space between P Tag ignore by browser)
The default font size for paragraphs (p) is equal to the **base size (1em)**, typically **16px**.

```
<html>
<body>
  <p>This is a paragraph.</p>
</body>
</html>
```

Output:

This is a paragraph.

- **<pre> Tag:** Defines **preformatted text**. Text in a <pre> element is displayed in a fixed-width font, and the text preserves both spaces and line breaks. The text will be displayed exactly as written in the HTML source code.

```
<html>
<body>
  <pre>
Text in a pre element
is displayed in a fixed-width
font, and it preserves
both    spaces and
line breaks
  </pre>
</body>
</html>
```

Output:

```
Text in a pre element
is displayed in a fixed-width
font, and it preserves
both    spaces and
line breaks
```

<pre> (Preformatted Text):

- Preserves **whitespace**, **line breaks**, and **indentation** as they appear in the HTML code.
- Displays text in a **monospace font** by default.
- Ideal for displaying code snippets or text that needs to retain its original formatting.

<p> (Paragraph):

- Treats multiple spaces, tabs, and line breaks as a **single space**.
- Used for defining a paragraph of text with **automatic line wrapping**.
- Does not preserve the whitespace or line breaks in the source code.

HTML Heading: <h1> to <h6>

HTML headings are titles or subtitles that you want to display on a webpage. HTML headings are defined with the <h1> to <h6> tags. <h1> defines the most important heading. <h6> defines the least important heading.

<h1> headings should be used for main headings, followed by <h2> headings, then the less important <h3>, and so on.

```
<html>
<body>
  <h1>Heading 1</h1>
  <h2>Heading 2</h2>
  <h3>Heading 3</h3>
  <h4>Heading 4</h4>
  <h5>Heading 5</h5>
  <h6>Heading 6</h6>
</body>
</html>
```

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

Tip: Use h1 to h6 elements only for headings. Do not use them just to make text bold or big. Use other tags for that.

The default font size of headings from the largest (h1) to the smallest (h6) is as follows (Ref*)

- h1 has a font size that is **2em**. 2 times the base size, equivalent to 32px if the base font size is 16px.
- h2 has a font size that is **1.5em**. 1.5 times the base size, which is 24px.
- h3 has a font size that is **1.17em**. 1.17 times the base size, equal to approximately 18.72px.
- h4 has a font size that is **1em**, which is equal to the base size, which is 16px.
- h5 has a font size that is **0.83em**. 0.83 times the base size, which is approximately 13.28px.
- h6 has a font size that is **0.67em**. 0.67 times the base size, which is approximately 10.72px.

This is based on the assumption that the base font size of the html or body is 16px, and adjusting this base size will proportionally affect all heading sizes.

Data Formatting tags

1. ****: Defines **bold text**

```
<body>
<p>This text is normal.</p>
<p><b>This text is bold.</b></p>
</body>
```

Output:

This text is normal.
This text is bold.

2. **<i>**: Defines a part of text in an alternate voice or mood. The content inside is typically displayed in *italic*.

```
<body>
<p>This text is normal.</p>
<p><i>This text is italic.</i></p>
</body>
```

Output:

This text is normal.
This text is italic.

3. **<u>**: Content inside is typically displayed with an **underline**.

```
<body>
<p> <u>Write Your First Paragraph in underlined text.</u></p>
</body>
```

Output:

Write Your First Paragraph in underlined text

4. ****: Define emphasized text. The content inside is typically displayed in *italic*.

```
<body>
  <em>This sentence has emphasized meaning.</em>
</body>
```

Output:

This sentence has emphasized meaning.

Note: Despite the fact that both `<em>` and `<i>` tags give the same visual effect, i.e. italic text, search engine robots pay more attention to the first one. That's why it is recommended to use the `<em>` tag for **website optimization**. `<em>` tag informs Google crawlers that the selected content is particularly important and deserves attention.

Same for the `<strong>` and `<b>` tags. Both tags give the same visual effect, but `<strong>` tag informs Google crawlers that the selected content is particularly important and deserves attention.

5. `<strong>`: Define text with strong importance. The content inside is typically displayed in **bold**.

```
<body>
<p>This text is normal.</p>
<p><strong>This text is important!</strong></p>
</body>
```

Output:

This text is normal.

This text is important!

6. `<br>`: It inserts a single line break. It is an empty tag which means that it has no end tag.

```
<p>This is sample example</p>
<p>This is <br> sample example</p>
```

Output:

This is sample example

This is
sample example

7. `<hr>`: Stands for horizontal rule and is used to insert a horizontal rule or a thematic break in an HTML page to divide or separate document sections.

The `<hr>` tag is an empty tag, and it does not require an end tag.

```
<h3>Textt1</h3>
<hr size="10" width="40%">
<h3>Textt2</h3>
<hr size="20" width="50%" noshade align="right">
<h3>Textt3</h3>
<hr size="2">
```

Output:

Textt1

Textt2

Textt3

- 8. <sup>:** Defines the superscript text. Superscript text appears half character above the normal line and is sometimes rendered in a smaller font. Superscript text can be used for footnotes.

```
<h3>logx<sup>2</sup></h3>
```

Output:

logx²

- 9. <sub>:** Defines the subscript text. Subscript text appears half a character below the normal line and is sometimes rendered in a smaller font.

```
<h4>CH<sub>3</sub></h4>
```

```
<h4>H<sub>2</sub>SO<sub>4</sub></h4>
```

Output:

CH₃

H₂SO₄

- 10. <ins>:** The `<ins>` tag defines a text that has been inserted into a document. Browsers will usually underline inserted text.

- 11. :** The `<del>` tag defines text that has been deleted from a document. Browsers will usually strike a line through deleted text.

```
<h1>My favorite color is <del>blue</del> <ins>red</ins>!</h1>
```

Output:

My favorite ~~color~~ is ~~blue~~ red!

Note: `<u>` is purely for **visual underlining** without meaning. `<ins>` is for **indicating added or inserted content**, with underlining being the default style. In modern web development, it's generally better to use semantic tags like `<ins>` when conveying meaning and `<u>` only when you want underlining purely for design.

- 12. <mark>:** define the marked text. It is used to highlight the part of the text in a paragraph. By default, `<mark>` tag highlight the text content in yellow color.

```
<h1>Here is your <mark>Highlighted text</mark></h1>
```

Output:

Here is your Highlighted text

HTML Link: Anchor tag (<a>)

Defines a hyperlink, which is used to link from one page to another. The most important attribute of the <a> element is the href attribute, which indicates the link's destination.

Attributes:

- ✓ **href:** Specifies the URL of the page the link goes to
- ✓ **target:** Specifies where to open the linked document
- ✓ **name:** is used to specify the name for an <a> element. It is used to reference the form data after submitting the form or to reference the element in a JavaScript.

```
<a href="url" target="_blank" | _self" id="a_link">Test</a>
```

_blank = Opens the linked document in a new window or tab

_self = Opens the linked document in the same frame as it was clicked (this is default)

Example:

```
<a href="https://www.example.com" target="_blank">Click here</a>
```

- ✓ When rendered in a browser, it would look like this: [Click here](https://www.example.com)
- ✓ If you click it, it will open <https://www.example.com> in a new tab.
- ✓ When target="_blank" is used, it tells the browser to open the link in a **new browser tab** or window (depending on browser settings).
- ✓ This is helpful when you want to keep the current page open while navigating to another site.

Example: Create a webpage that contains a long block of text and a link at the bottom which, when clicked, scrolls the user back to the top of the page. The solution should demonstrate how to use the id attribute to define an anchor point () and the href="#..." link to navigate back to that point.

```
<html>
  <body>
    <p id="test"></p>
    <h1>
      <pre>
        a
        b
        c
        d
        e
        f
        g
        h
        i
        j
        k
        l
        m
        n
```



```
o
p
q
r
s
t
u
v
w
x
y
z
a
b
c
d
e
f
g
h
i
j
</pre>
</h1>
<a href="#test">Scroll to top</a>
</body>
</html>
```

1. ****: A unique identifier used to reference or target elements. It is the preferred method in modern HTML. Using id is more current and widely supported.

2. **Scroll to top**

This creates a hyperlink that, when clicked, will scroll the user back to the point in the page where the named anchor (****) is located. The **#test** in the **href** attribute reference.

Note:

name: Used in older versions of HTML for named anchors, but mostly for form elements now. It's deprecated for anchor elements in HTML5.

3. **id**: A unique identifier used to reference or target elements. It is the preferred method in modern HTML. In the example provided, the functionality would remain the same if you replaced **name="test"** with **id="test"**, but using id is more current and widely supported.

❖ id and class attributes

• id attribute

- A unique identifier for a single element on a page.
- Used to **target that element** in CSS or JavaScript.
- Each id must be unique (only one per page).

• class attribute

- A reusable label that can be assigned to multiple elements.
- Used to **apply the same styles or behaviors** to groups of elements.

Why use them?

- To **style elements** with CSS.
- To **manipulate elements** with JavaScript.
- To provide **structure and meaning** in HTML for easier maintenance and reusability.

In short:

- **id = unique identifier (one element).**
- **class = reusable category (many elements).**

Customize Link color without CSS (Ref*)

```
<html>
<body link="blue" alink="green" vlink="yellow">
<p><a href="https://www.yahoo.co.in" target="_blank">yahoo</a></p>
</body>
</html>
```

TAGS – Summary

Category	Tag	Description
Layout & Sectioning Tags	<div>	Generic container, used for grouping elements.
		Inline container for styling a small portion of text.
	<header>	Represents the top section of a webpage or article.
	<footer>	Bottom section of a webpage (credits, copyright).
	<main>	Main content of the webpage.
	<section>	Defines sections in a webpage.
	<article>	Represents self-contained content (e.g., blog post).
	<aside>	Sidebar content, not the main focus.
	<nav>	Navigation links/menu.
Content & Text Tags	<h1> to <h6>	Headings (h1 is largest, h6 is smallest).
	<p>	Paragraph of text.
	<pre>	Preformatted text (preserves spaces, line breaks).
	 	Line break.
	<hr>	Horizontal line (divider).
Hyperlink Tag	<a>	Anchor tag used for hyperlinks.
Text Formatting Tags		Bold text (visual only).
		Important text (semantic meaning).
	<i>	Italic text (visual only).
		Emphasized text (semantic meaning).
	<u>	Underlined text.
	<mark>	Highlighted text.
		Deleted/strikethrough text.
	<ins>	Inserted/underlined text.
	<sub>	Subscript (below text).
	<sup>	Superscript (above text).
	<abbr>	Abbreviation (shows full form on hover).

Example:

```
<html>
<head>
  <title>HTML Tags Example</title>
</head>
<body>
  <!-- Layout & Sectioning Tags -->
  <header>Website Header</header>
  <nav>Navigation Bar</nav>
```

```

<main>
  <section>
    <article>
      <h1>Main Heading</h1>
      <p>This is a <b>bold</b> word and this is <i>italic</i>.</p>
      <p>You can also <strong>highlight importance</strong> or <em>emphasize text</em>.</p>
      <p>
        Here is <u>underlined</u>, <mark>highlighted</mark>, <del>deleted</del>,
        and <ins>inserted</ins> text.
      </p>
      <p>Water formula: H<sub>2</sub>O and Square: x<sup>2</sup></p>
      <p>
        <abbr title="HyperText Markup Language">HTML</abbr> is the base of web
        pages.
      </p>
      <pre>
        Preformatted text keeps
        spaces & line breaks.
      </pre>
      <p>Line break here<br>Next line</p>
      <hr>
      <a href="https://example.com" target="_blank">Visit Example.com</a>
    </article>
  </section>
<aside>Sidebar Information</aside>
</main>
<footer>Website Footer</footer>
</body>
</html>

```

Output:

Website Header
Navigation Bar

Main Heading

This is a **bold** word and this is *italic*.

You can also **highlight importance** or *emphasize text*.

Here is underlined, **highlighted**, ~~deleted~~, and inserted text.

Water formula: H₂O and Square: x²

HTML is the base of web pages.

```

Preformatted text keeps
spaces & line breaks.

```

Line break here
Next line

[Visit Example.com](https://example.com)
Sidebar Information
Website Footer

HTML Lists

A list is a record of short pieces of related information or used to display the data or any information on web pages in the ordered or unordered form. For instance, to purchase the items, we need to prepare a list that can either be ordered or unordered list which helps us to organize the data & easy to find the item.

There are two types of lists:

- 1. Unordered list:** An unordered list starts with the `<ul>` tag. Each list item starts with the `<li>` tag. The list items will be marked with disc (small black circles) by default.

Attribute: type = "disc/square/circle/none"

```
<ul>
  <li>Test 1</li>
  <li>Test 2</li>
  <ul type="square">
    <li>Test 3</li>
    <li>Test 4</li>
  </ul>
</ul>
```

- Test 1
- Test 2
 - Test 3
 - Test 4

Output:

type="disc":

Items are marked with **filled circles** as bullet points. This is default.

- Item 1
- Item 2
- Item 3

type="square":

Items are marked with **squares**.

- Item 1
- Item 2
- Item 3

type="circle":

Items are marked with **hollow circles**.

- Item 1
- Item 2
- Item 3

type="none":

No bullet points are shown for the list items.

- Item 1
- Item 2
- Item 3

2. Ordered list: An ordered list starts with the **** tag. Each list item starts with the **** tag. The list items will be marked with numbers by default:

Attribute:

type: type=" 1/ i/ I/ a/ A"

start: Specifies the start value of an ordered list

reversed: Specifies that the list order should be reversed (9,8,7...)

```
<ol start="5">
  <li>Test 1</li>
  <li>Test 2</li>
  <li>Test 3</li>
  <ol type="A">
    <li>Test 4</li>
    <li>Test 5</li>
    <ol reversed type="i">
      <li>Test 6</li>
      <li>Test 6</li>
    </ol>
  </ol>
</ol>
```

Output:

```
5. Test 1
6. Test 2
7. Test 3
  A. Test 4
  B. Test 5
    ii. Test 6
    i. Test 6
```

For type 'a' 27,28,29.. letters are aa,ab,ac... respectively.

Nested List Example:

```

<body>
  <ol type="A" start="4">
    <li>List1</li>
    <li>List2</li>
    <ul type="square">
      <li>list2.1</li>
      <li>list2.2</li>
      <ol type="I" reversed >
        <li>list 2.2.1</li>
        <li>list 2.2.1</li>
        <li>list 2.2.1</li>
      </ol>
    </ul>
    <li>List3</li>
    <ul type="none">
      <li>list 3.1</li>
      <li>list 3.2</li>
    </ul>
    <li>list4</li>
    <ol type="a">
      <li>list 4.1</li>
      <ul type="circle">
        <li>list 4.1.1</li>
        <li>list 4.1.2</li>
      </ul>
      <li>list 4.1</li>
    </ol>
  </ol>
</body>

```

Output:

```

D. List1
E. List2
  ■ list2.1
  ■ list2.2
    III. list 2.2.1
    II. list 2.2.1
    I. list 2.2.1
F. List3
  list 3.1
  list 3.2
G. list4
  a. list 4.1
    ○ list 4.1.1
    ○ list 4.1.2
  b. list 4.1

```

Media Tags

❖ Tag

- Used to display images.
- Attributes: src (image source), alt (alternative text), width, height.

```

```

<figure> and <figcaption>

- <figure> groups media content (like images or videos) with an optional caption.
- <figcaption> provides a caption for the media inside <figure>.

Example: Understand the concept of ,<figure>,<figcaption>.

```

```

```
<figure>
```

```
  
```

```
  <figcaption>Html Image</figcaption>
```

```
</figure>
```



Using <figure> with is useful because it **semantically groups an image with its caption**, making the content more meaningful and accessible. The <figcaption> inside <figure> provides a clear description or context for the image, which improves **accessibility, SEO, and content clarity**.

In short: **<figure> + = image + optional caption, grouped meaningfully.**

Information:**Absolute File Paths**

An absolute file path is the full URL to a file:

```

```

Relative File Paths

A relative file path points to a file relative to the current page.

In the following example, the file path points to a file in the images folder located at the root of the current web:

```

```

** tag with <a> tag**

To use image as a link in HTML, use the tag as well as the <a> tag with the href attribute.

The tag is for using an image in a web page and the <a> tag is for adding a link. Under the image tag src attribute, add the URL of the image. With that, also add the height and width

- ✓ Always use the alt attribute inside the tag to provide descriptive text for accessibility and better SEO.

Example:

```
<body>
<h1>The img element</h1>
<a href="https://www.google.com/">

</a>
</body>
```

Output:

The img element



❖ **<audio> Tag**

- Embeds audio files.

Syntax:

```
<audio>
<source src="sample.mp3" type="audio/mp3">
</audio>
```

***For single source:** `<audio src="sample.mp3" type="audio/mp3" controls>`

Attributes of <audio> tag:

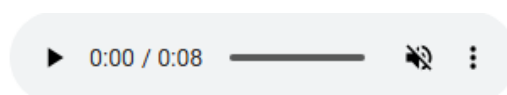
controls	Specifies that audio controls should be displayed (such as a play/pause button etc) Eg. <code><audio controls></code> (Must Include)
autoplay	Specifies that the audio will start playing as soon as it is ready. Eg. <code><audio controls autoplay></code>
loop	Specifies that the audio will start over again, every time it is finished. Eg. <code><audio controls loop></code>
muted	Specifies that the audio output should be muted. (browsers allow muted autoplay). Eg. <code><audio controls muted></code>

Attributes of <source> tag:

src	Designates the URL of the audio file.
type	It contains single value MIME_type which specifies the MIME_type of the media resource. Eg. audio/mp3, audio/mpeg, audio/wav, audio/ogg

Example:

```
<audio controls autoplay muted loop>
<source src="1.mp3" type="audio/mpeg">
Your browser does not support the audio element.
</audio>
```

**To allow play audio automatically on muted mode**

1. Open Chrome.
2. Go to **Settings** → **Privacy and Security** → **Site Settings**.
3. Scroll to **Additional content settings** → **Sound**.
4. Here you can:
 - Allow all sites to play sound automatically, OR
 - Add specific sites to **Allow** list (so they can autoplay with sound).

❖ **<video> Tag**

- Embeds video files.

Attributes of <video> tag:

controls	Specifies that video controls should be displayed (such as a play/pause button etc) Eg. <code><video controls></code>
autoplay	Specifies that the video will start playing as soon as it is ready. Eg. <code><video controls autoplay></code> **Add muted after autoplay to let your video start playing automatically (but muted):
loop	Specifies that the video will start over again, every time it is finished. Eg. <code><video controls loop></code>
muted	Specifies that the video output should be muted. Eg. <code><video controls muted></code>
src	Designates the URL of the video file.
height	Sets the height of the video player in pixels.
width	Sets the width of the video player in pixels.
Poster (*ref)	Path of an image to be shown while the video is downloading, or until the user hits the play button.

Attributes of <source> tag:

src	Designates the URL of the video file.
type	It contains single value MIME_type which specifies the MIME_type of the media resource. Eg. video/mp4, video/ogg

Example:

```
<video autoplay controls muted loop width="700" height="500">
  <source src="2.mp4" type="video/mp4" >
  Your browser does not support the video element.
</video>
```



<source>

- Specifies multiple media sources for <audio> or <video> so the browser can choose a supported format.

Example:

```
<video width="400" controls>
  <source src="video.mp4" type="video/mp4">
  <source src="video.ogv" type="video/ogg">
</video>
```

❖ <iframe> Tag

- Embeds another webpage, map, video (like YouTube), pdf etc inside your page.

Attribute	Purpose	Example from your code
src	URL or file path of embedded resource	src="https://www.wikipedia.org"
width	Sets width in pixels	width="600"
height	Sets height in pixels	height="400"
frameborder	Shows/hides border (deprecated)	frameborder="0"
style	Inline CSS (styling)	style="border:0;"
allowfullscreen	Allows fullscreen (mainly for video)	allowfullscreen (YouTube iframe)
loading	Lazy/eager load for performance	loading="lazy" (Google Maps iframe)
title	Accessibility/SEO description	title="YouTube video player"

Example:

```
<body>
  <h1>Iframe Examples</h1>

  <!-- 1. Another Webpage -->
  <h2>Embed Webpage</h2>
  <iframe src="https://www.wikipedia.org" width="600" height="400" frameborder="0"></iframe>

  <!-- 2. YouTube Video -->
  <h2>Embed YouTube Video</h2>
```

```
<iframe width="560" height="315"
src="https://www.youtube.com/embed/qz0aGYrrlhU?si=eNLgc597jsbAK9-2" title="YouTube video
player" frameborder="0" allowfullscreen></iframe>
```

```
<!-- 3. Google Maps -->
```

```
<h2>Embed Google Maps</h2>
```

```
<iframe
```

```
src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d3672.8940065186866!2d72.48528877
603572!3d22.990924817507025!2m3!1f0!2f0!3f0!3m2!1i1024!2i768!4f13.1!3m3!1m2!1s0x395e9aee6153
92b1%3A0x30d4b8160142adf3!2sL%20J%20Institute%20Of%20Management%20Studies!5e0!3m2!1sen!2
sin!4v1756790030904!5m2!1sen!2sin" width="600" height="450" style="border:0;" allowfullscreen=""
loading="lazy"></iframe>
```

```
<!-- 4. PDF Document -->
```

```
<h2>Embed PDF</h2>
```

```
<iframe src="Chapter 1-Introduction.pdf" width="600" height="500"></iframe>
```

```
<!-- 5. Local HTML File -->
```

```
<h2>Embed Local Page</h2>
```

```
<iframe src="fv.html" width="500" height="300" frameborder="1"></iframe>
```

```
</body>
```

Iframe Examples

Embed Webpage



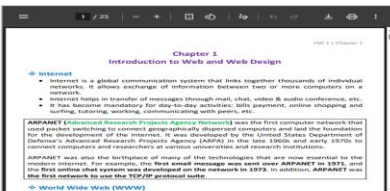
Embed YouTube Video



Embed Google Maps



Embed PDF

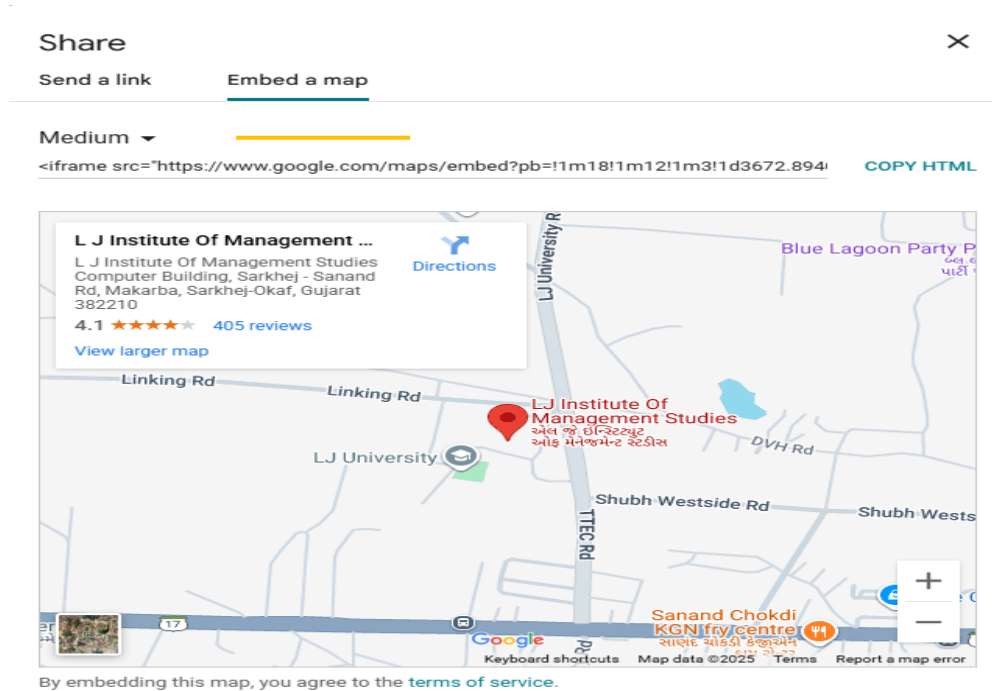
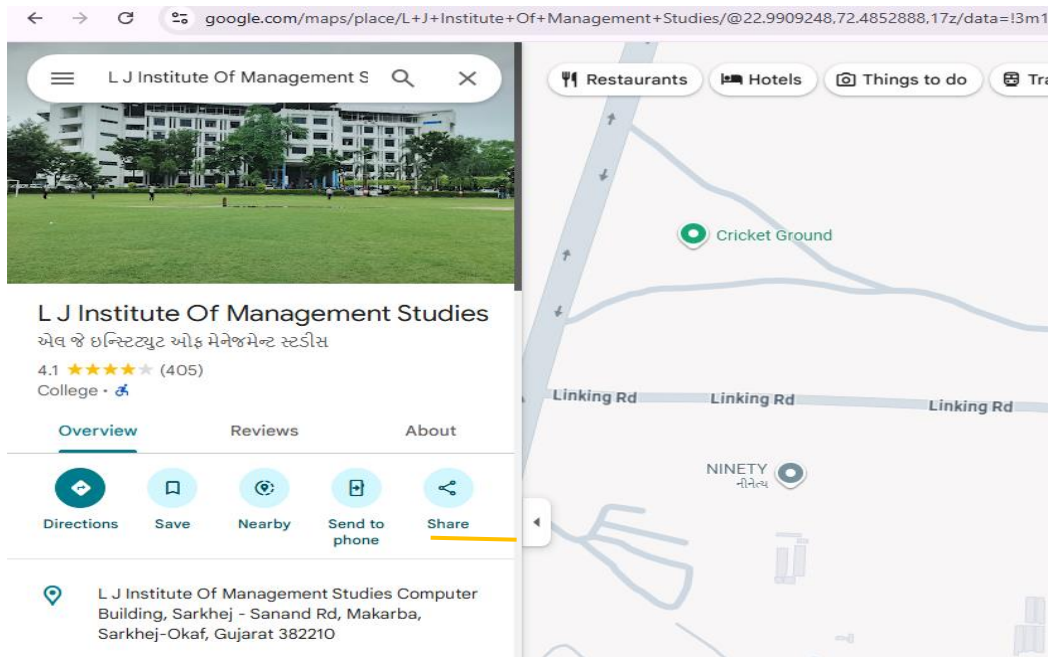


Embed Local Page



Embed Map – Open Google Map

- 1) Go to Google Maps.
- 2) Search for the location you want.
- 3) Click the **Share** button.
- 4) Choose the **Embed a map** tab.
- 5) Copy the `<iframe>` code provided.



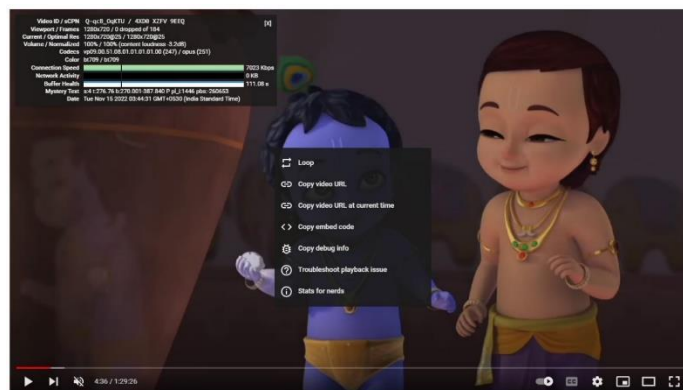
Embed a YouTube Video

Method-1 Steps to get complete <iframe> html element

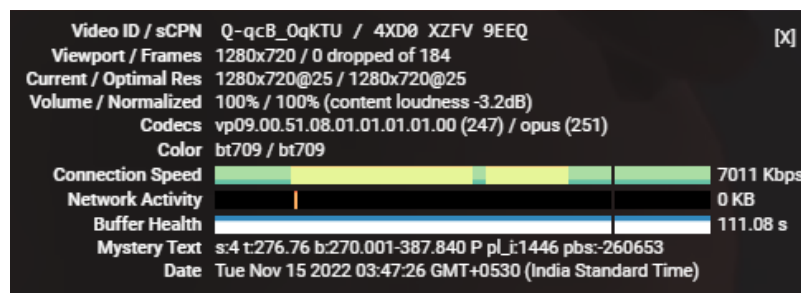
- 1) Go to the YouTube video you want to embed.
- 2) Click the **Share** button (below the video).
- 3) Select **Embed**.
- 4) Copy the <iframe> code that YouTube generates.

Method-2 Steps to get the only Video Id of a YouTube video:

- Open the YouTube video whose Id you want.
- Right click on the video, from the menu select “Stats for nerds”.



- The first value in the box is the Video ID. Same can found from URL.



- To embed youtube video `src="https://www.youtube.com/embed/Q-qcB_OqKTU"`

embed = An embedded video lets you borrow the video from another platform. Visitors can watch the video on your website without leaving the current page.

video id = Q-qcB_OqKTU

API parameters for Youtube video

Name	Description
controls	Specifies that youtube video controls should be displayed (such as a play/pause button etc) Value 0: Player controls does not display.
	Value 1 (default): Player controls display.

	Eg. <iframe width="500" height="500" src="https://www.youtube.com/embed/ Q-qcB_OqKTU?controls=0"> </iframe>
autoplay	Specifies that the youtube video will start playing as soon as it is ready.
mute	Specifies that the youtube video output should be muted. Eg. <iframe width="500" height="500" src="https://www.youtube.com/embed// Q-qcB_OqKTU?autoplay=1&mute=1"> </iframe>
loop	Specifies that the youtube video will start over again, every time it is finished. Eg. <iframe width="500" height="500" src="https://www.youtube.com/embed/Q-qcB_OqKTU?playlist=Q-qcB_OqKTU&loop=1"> </iframe>
src	Designates the URL of the youtube video file.
height	Sets the height of the youtube video player in pixels.
width	Sets the width of the youtube video player in pixels.
title="YouTubevideoplaylist": This provides a title for the iframe, which is useful for accessibility and SEO purposes. The title helps screen readers and search engines understand the content of the embedded video.	
allowfullscreen: This attribute allows users to view the video in fullscreen mode by clicking the fullscreen button (if visible in the embedded player)	

Example 1:

```
<iframe width="500" height="500" src="https://www.youtube.com/embed/Q-qcB_OqKTU?&playlist=Q-qcB_OqKTU&mute=1&autoplay=1&loop=1">
</iframe>
```


HTML Table

- The `<table>` tag defines HTML table.
- An HTML table consists of one `<table>` element and one or more `<tr>`, `<th>`, and `<td>` elements.
- The `<tr>` element defines a table row, the `<th>` element defines a table header, and the `<td>` element defines a table cell.
- An HTML table may also include `<caption>`, `<thead>`, `<tfoot>`, and `<tbody>` elements.

HTML tags for table:

<code><caption></code>	The <code><caption></code> tag defines a table caption. The <code><caption></code> tag must be inserted immediately after the <code><table></code> tag. Tip: By default, a table caption will be center-aligned above a table. However, the CSS properties <code>text-align</code> and <code>caption-side</code> can be used to align and place the caption.
<code><table></code>	Defines a table.
<code><th></code>	Defines a header cell in a table
<code><tr></code>	Defines a row in a table
<code><td></code>	Defines a cell in a table
<code><thead></code>	Groups the header content in a table
<code><tbody></code>	Groups the body content in a table
<code><tfoot></code>	Groups the footer content in a table

HTML `<table>` tag attributes:

Attribute	Value	Description
<code>align</code>	right left center justify char	Deprecated – Visual alignment.
<code>bgcolor</code>	<code>rgb(x,x,x)</code> <code>#hexcode</code> colorname	Deprecated – Specifies the backgroundcolor of the table.
<code>border</code>	pixels	Deprecated – Specifies the border width.A value of "0" means no border.
<code>cellpadding</code>	pixels or %	Deprecated – Specifies the space between the cell borders and their contents.

cellspacing	pixels or %	Deprecated – Specifies the space between cells.
rules	none, groups, rows, cols, all	Deprecated – The HTML <table> rules Attribute is used to <i>specify which parts of the inside borders that should be visible</i>. Syntax: <table rules="value"> Attribute Values: • none: It does not create any lines. • groups: It create lines between row and column groups. • rows: It creates line between the rows. • cols: It creates line between the columns. • all: It creates line between the rows and columns. Note: The <table> rules Attribute is not supported by HTML 5.
width	pixels or %	Deprecated .Specifies the width of the table.

<td><th><tr> tags attributes:

Attribute	Value	Description
align	right left center justifychar	Deprecated – Visual alignment.
bgcolor	rgb(x,x,x) #hexcode colorname	Deprecated – Specifies the backgroundcolor of the cell.
colspan	Number of columns to merge (use in <td> or <th>)	Number of columns a header cell shouldspan
rowspan	Number of rows to merge (use in <td> or <th>)	Set the number of rows a header cell shouldspan.

Example 1

Table

Name	Cost
A	20
B	30
Total	50

```
<table border="4" cellpadding="10">
  <caption>Table</caption>
  <thead>
    <tr>
      <th>Name</th>
      <th>Cost</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>A</td>
      <td>20</td>
    </tr>
    <tr>
      <td>B</td>
      <td>30</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <th>Total</th>
      <td>50</td>
    </tr>
  </tfoot>
</table>
```

Example 2

C	A
	b

```
<table border="2" cellspacing="4" cellpadding="30">
  <tr>
    <td rowspan="2">C</td>
    <td>A</td>
  </tr>
  <tr>
    <td>b</td>
  </tr>
</table>
```

Example3:

A	
B	C

```
<table border="2" rules="rows" cellpadding="30">
  <tr>
    <td colspan="2" align="center">A</td>
  </tr>
  <tr>
    <td>B</td>
    <td>C</td>
  </tr>
</table>
```

Example 4

Header spanning 2 columns		Rowspan Header
Cell 1	Cell 2	
Row 2, Col 1	Merged across 2 columns	

```

<!DOCTYPE html>
<html>
<head>
  <title>Deprecated Table Attributes Example</title>
</head>
<body>

<!-- Using deprecated attributes just for demonstration -->
<table align="center" bgcolor="lightyellow" border="2" cellpadding="10" cellspacing="5" rules="all"
width="80%">

  <tr bgcolor="lightblue">
    <th align="left" colspan="2">Header spanning 2 columns</th>
    <th rowspan="2" bgcolor="lightyellow">Rowspan Header</th>
  </tr>

  <tr bgcolor="lightpink">
    <td align="center">Cell 1</td>
    <td align="right">Cell 2</td>
  </tr>

  <tr bgcolor="gray">
    <td>Row 2, Col 1</td>
    <td colspan="2" align="justify">Merged across 2 columns</td>
  </tr>
</table>
</body>
</html>

```

- ✓ If you want **visible grid lines** → use rules="all" and set cellspacing="0".
- ✓ If you want **spacing between cells** → use cellspacing, but then the effect of rules is lost.

Example 5

Table					
A	B	D	E		F
	C		G	H	I
J					
K			L	M	

```
<html>
<head><title> Table </title></head>
<body align='center'>
  <table border='3px' width='50%' align="center" cellpadding='15' cellspacing="5">
    <caption><strong><font size="6"> Table</font></strong></caption>
    <tr>
      <td rowspan='2' bgcolor="yellow"> A</td>
      <td bgcolor="lightgreen"> <b>B</b> </td>
      <td rowspan='3' bgcolor="yellow"> D </td>
      <td colspan='2' bgcolor="pink"> E </td>
      <td bgcolor="lightblue"> F </td>
    </tr>
    <tr>
      <td bgcolor="lightblue"> C </td>
      <td rowspan='2' bgcolor="lightblue"> G </td>
      <td rowspan='2' bgcolor="lightgreen"> <b>H</b> </td>
      <td rowspan='2' bgcolor="yellow"> I </td>
    </tr>
    <tr>
      <td colspan='2' bgcolor="pink"> J </td>
    </tr>
    <tr>
      <td colspan='3' bgcolor="lightgreen"> <b>K</b> </td>
      <td bgcolor="yellow"> L </td>
      <td colspan='2' bgcolor="pink"> M </td>
    </tr>
  </table>
</body></html>
```

Example 6

Design an HTML table for **LJ University - Student List** that displays the **Branch, Student Name, and Roll Number** of students.

The table should:

1. Use the **rules="groups"** attribute so that horizontal lines are drawn **only between row groups** (different branches).
2. Include at least **two different branches** (e.g., Computer Science, Mechanical).
3. Each branch should have **multiple students**, and the **branch name should span multiple rows** using the rowspan attribute.
4. Add a **table header** (<thead>) with column titles (Branch, Student Name, Roll Number).
5. Add a **footer row** (<tfoot>) with copyright information for LJ University.
6. Show background color for different row groups (<tbody>) so that each branch is visually distinct.

```
<!DOCTYPE html>
<html>
<head>
  <title>College Table - Rules Groups Example</title>
</head>
<body>

<h2 align="center">LJ University - Student List</h2>

<table border="2" rules="groups" cellspacing="0" cellpadding="8" align="center" width="50%">

  <thead bgcolor="#ddd">
    <tr>
      <th>Branch</th>
      <th>Student Name</th>
      <th>Roll Number</th>
    </tr>
  </thead>

  <tbody bgcolor="lightblue">
    <tr>
      <td rowspan="3">Computer Science</td>
      <td>Rahul Sharma</td>
      <td>CS101</td>
    </tr>
    <tr>
      <td>Priya Patel</td>
      <td>CS102</td>
    </tr>
    <tr>
      <td>Amit Verma</td>
      <td>CS103</td>
    </tr>
  </tbody>
```

```
<tbody bgcolor="lightyellow">
  <tr>
    <td rowspan="3">Mechanical</td>
    <td>Rohan Mehta</td>
    <td>ME201</td>
  </tr>
  <tr>
    <td>Sneha Iyer</td>
    <td>ME202</td>
  </tr>
  <tr>
    <td>Vikas Sharma</td>
    <td>ME203</td>
  </tr>
</tbody>

<tfoot bgcolor="#ddd">
  <tr>
    <td colspan="3" align="center">LJ University © 2025</td>
  </tr>
</tfoot>
</table>

</body>
</html>
```

LJ University - Student List

Branch	Student Name	Roll Number
Computer Science	Rahul Sharma	CS101
	Priya Patel	CS102
	Amit Verma	CS103
Mechanical	Rohan Mehta	ME201
	Sneha Iyer	ME202
	Vikas Sharma	ME203
LJ University © 2025		

HTML Block and Inline elements: <div> and [Reference Only*]

❖ <div> Tag

The div tag is known as Division tag. The div tag is used in HTML to make divisions of content in the web page like (text, images, header, footer, navigation bar, etc). Div tag has both open(<div>) and closing (</div>) tag and it is mandatory to close the tag.

It is a block level tag. This defines specific section or division in html page. This gives style to entire block and all-inclusive contents.

It is used to the group of various tags of HTML so that sections can be created and style can be applied to them.

- Div tag is Block level tag
- It is a generic container tag
- It is used to the group of various tags of HTML so that sections can be created and style can be applied to them.

Example:

```
<html>
<body>
  <div style="background-color:lightblue;text-align:center;">
    <h1>This is heading in div element. </h1>
    <p>This is some text in div element. </p>
  </div>
  <p>This is some text outside of div element. </p>
</body>
</html>
```

Output:

This is heading in div element.

This is some text in div element.

This is some text outside of div element.

❖ Tag

The HTML span element is a generic **inline container** for inline elements and content. It is used to group elements for styling purposes (by using the class or id attributes), A better way to use it when no other semantic element is available.

The span tag is a paired tag means it has both open (<) and closing (>) tags, and it is mandatory to close the tag. The span tag is used for the grouping of inline elements & this tag does not

make any visual change by itself. span is very similar to the div tag, but div is a block-level tag and span is an inline tag and takes only required Width. (div tag: Contain Whole Width Available)

Example:

```
<html>

<body>

  <h1>Example of span </h1>
  <p>
    Full
    <span style="color:cyan;font-weight:bolder">
      Stack
    </span>
    <span style="background-color: lightcoral;">
      Development
    </span>
  </p>
</body>

</html>
```

Output:

Example of span

Full **Stack** Development

Difference between span and div tag

Properties	Div Tag	Span Tag
Elements Types	Block-Level	Inline
Space/Width	Contain Whole Width Available	Takes only required Width
Examples	Headings, Paragraph, form	Attribute, image
Uses	Web-layout	container for some text

Task: Do as Direct (Reference *)

```

<html>
<head>
  <title>Document</title>
</head>
<body>
  <div align="center" style = "background-color:orange;height:100px">
    <h1>Div Tag</h1>
    <p>This section is written inside Div using basic inline css property</p>
  </div>
  <div align="center" style = "background-color:white;height:50px;color:blue">
    <p></p>
  </div>
  <div align="center" style = "background-color:green;height:100px">
    <h2>This section is written inside third Div using basic inline css property</h2>
  </div>
  <br>
  <br>
  <br>
  <h2 align="center"> Span Tag</h2>
  <h3 align="center">To give inline effect use span tag for example:
    <span style="color:red;background-color: rgb(156, 78, 150);">R</span>
    <span style="color:orange;background-color: rgb(156, 78, 152)">A</span>
    <span style="color:yellow;background-color: rgb(156, 78, 152)">I</span>
    <span style="color:green;background-color: rgb(156, 78, 152)">N</span>
    <span style="color:blue;background-color: rgb(156, 78, 152)">B</span>
    <span style="color:indigo;background-color: rgb(156, 78, 152)">O</span>
    <span style="color:violet;background-color: rgb(156, 78, 152)">W</span>
  </h3>
</body>
</html>

```

Output:**Div Tag**

This section is written inside Div using basic inline css property



This section is written inside third Div using basic inline css property

Span Tag

To give inline effect use span tag for example: RAINBOW